

Wenrui XU

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EDUCATION

Ph.D. Student, Environmental Planning, Dept. of Landscape Arch. & Envi. Planning 2025-2029(Expected)
UNIVERSITY OF CALIFORNIA, BERKELEY CA, USA

- GPA: 4.00/4.00
- *Advisor*: Prof. Lu Liang, Geospatial 3M(Monitoring-Mapping-Modeling) Lab
- *Coursework*: Remote Sensing of Environment (A+); Geographical Information Science (A); Applied Remote Sensing (A+); Principles and Techniques of Data Science (A+); Data Science for Social Change (A)

Master of Architecture, Urban Informatics Program, School of Architecture 2023-2025
TSINGHUA UNIVERSITY Beijing, China

- GPA: 3.93/4.00 | GRE: 336 (Verbal 166, Quantitative 170) | TOEFL: 107 (Reading 30, Listening 29, Writing 26)
- *Honor*: Comprehensive Excellence Scholarship
- *Main Courses*: Big Data & Urban Planning (A-); Urban Informatics (A-); Urban Renewal & Governance (A)
- *Certificate in Global Private Equity Research Institute*: Corporate Finance (A); Analyzing Financial Statements(A-); Private Equity and Venture Capital (A); Institutional Buy-side Investment in China (A-)
- *Graduation Project*: (Constructed) Zhejiang Wanli University Landscape Planning and Building Design

Bachelor of Engineering, Architecture and Urban Design, College of Arch. & Urban Planning 2019-2023
TONGJI UNIVERSITY Shanghai, China

- GPA: 4.89/5.00 | *Ranking*: 1% | CET-6: 684 | CET-4: 642
- *Honor*: Graduation with the Highest Distinction (Top 1% of all graduates in Tongji University and Shanghai) National Scholarship (Top 1%); Outstanding Student model (7/4300+ undergraduates) Outstanding Student Leader; The First Prize Scholarships; Enterprise Scholarships
- *Graduation Project*: Port-Oriented and Landscape Urbanism: Hong Kong-Shenzhen Border Urban Design

FUDAN UNIVERSITY Shanghai, China

- *Minor Program (27 credits)*: Macroeconomics (A); Microeconomics (A-); Accounting (A); Financial Market (A)

PUBLICATIONS

PEER-REVIEWED JOURNAL ARTICLES † Co-first Author; * Corresponding Author; Full list in [Google Scholar](#)

- [1] **Xu, W.**†, Lyu, D.†, Wang, W.†, Feng, J., Gao, C., & Li, Y.* (2025). Defining and evaluating visual language models' basic spatial abilities: A perspective from psychometrics. Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (volume 1: Long Papers) (pp. 11571–11590). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2025.acl-long.567>
- [2] **Xu, W.**, & Gu, X.* (2025). Spatiotemporal impacts of purpose-specific human mobility on air pollution: Evidence from taxi trajectories and interpretable machine learning. Sustainable Cities and Society, 126, 106411. <https://doi.org/10.1016/j.scs.2025.106411>
- [3] Gu, X., **Xu, W.**, Gong, C., & Liu, X.* (2025). City centers really lived up to the hype? Evidence from human perceptions of over 4000 communities in China. Cities, 166, 106278. <https://doi.org/10.1016/j.cities.2025.106278>
- [4] Wang, S., Zhao, Z., Xie, Y., Ma, M., Chen, Z., Wang, Z., Su, B., **Xu, W.**, & Li, T. (2024). Recent surge in public interest in transportation: Sentiment analysis of baidu apollo go using weibo data. arXiv. <https://doi.org/10.48550/arXiv.2408.10088>

WORKING PAPERS † Co-first Author; * Corresponding Author

- [1] Gu, X., Liu, Y*, **Xu, W.**, Qiu, W., Li, X., Chen, X., Lu, S., Qiu, W.* (2025-). Scalable quantification of 3D material albedo reveals consistent cooling mechanisms and unequal mitigation potential. Communications Earth & Environment (1st Revision).
- [2] **Xu, W.**, Lan, Y., Yang, T., Chen, J.*, Chen, J., Qian, J., Han, W., Edle von Schickh, A. (2025-). Campus squares are significantly associated with students' emotions: A multimodal field experiment approach. Frontiers of Architectural Research (1st Revision).
- [3] **Xu, W.**, Liang, L.* (2025-). Compound heatwave-air pollution hazards across the United States. Manuscript in Preparation.
- [4] **Xu, W.**, Zhang, X., Sun, J., Liang, L.* (2026-). Fine-tuned vision-language models for wildfire risk assessment. Manuscript in Preparation.
- [5] **Xu, W.**, Agostini, G., Pierson, E.*, Blumenstock, J., Liang, L. (2026-). SAEarth: Can Sparse AutoEncoders expose interpretable features in Geospatial Foundation Models?. Manuscript in Preparation.
- [6] Lyu, D.†, **Xu, W.**†, Wang, W., Gao, C., Zhuang, W.*, Li, Y.* (2025-). Aesthetic perception of large language models. Manuscript in Preparation.
- [7] Lyu, D., Li, J., **Xu, W.**, Li, T.* (2025-). Spatial inequities in emergency rescue networks in China. Manuscript in Preparation.

PRESENTATIONS

- [2026] The Annual Meeting of the Association of American Geographers (AAG), Oral Presentation, San Francisco, USA.
- [2025] The 63rd Annual Meeting of the Association for Computational Linguistics (ACL), Online Oral Presentation, Vienna, Austria.
- [2025] The 65th Association of Collegiate Schools of Planning Annual Conference (ACSP), Minneapolis, USA.
- [2025] The 4th International Conference on Urban Informatics (GSCS & ICUI), Oral Presentation, Hong Kong, China.
- [2025] The 19th International Conference on Computational Urban Planning and Urban Management (CUPUM), Poster Presentation, London, UK.
- [2025] The 19th International Association for China Planning Annual Conference (IACP), Oral Presentation, Xiamen, China.
- [2025] The 32nd International Conference on Geoinformatics (CPGIS), Oral Presentation, Jiaozuo, China.
- [2025] The 85th Annual Meeting of the Academy of Management (AOM), Extending Knowledge-based View in Generative AI Era, Published in Proceedings, Copenhagen, Denmark.
- [2024] Nature Conference on Air Pollution and Climate Change, Poster Presentation, Beijing, China.

AWARDS AND HONORS

- [2026] Autodesk Graduate Student Fellowship (\$15,000)
- [2024] Comprehensive Excellence Scholarship (Tsinghua University)
- [2023] Distinct Graduate of Shanghai: Graduation with the highest distinction (Top 1%)
- [2023] Jundi Travel Grant (~\$30,000)
- [2022] The First Prize Undergraduate Scholarship (Tongji University)
- [2022] Exhibition of Architectural Design in Developing Countries: Bronze Award
- [2021] The First Prize Undergraduate Scholarship (Tongji University)
- [2021] Outstanding Student Model (Top 7 of 4300+ undergraduates at Tongji University, 0.1%)
- [2021] National Real Estate Innovation & Entrepreneurship Competition: Top Prize
- [2021] National Computer Design Competition for College Students: Second Prize (Top 1294/100,000+)
- [2020] National Scholarship: Highest honor for undergraduate (Top 1%)
- [2020] National English Competition for College Students: Top Prize (Top 23/43000+)
- [2018] China High School Mathematics League: Second Prize in Anhui (Top 681/499,000+)

PROFESSIONAL EXPERIENCE

Graduate Student Researcher, BAIR (Berkeley Artificial Intelligence Research Lab)

2026.04-

UNIVERSITY OF CALIFORNIA, BERKELEY

Berkeley, USA

- Leading a project on LLMs and foundation models.
- *Advisor:* Prof. Emma Pierson

Graduate Student Researcher, Geospatial 3M Lab

2026.01-

UNIVERSITY OF CALIFORNIA, BERKELEY

Berkeley, USA

- Leading a project on fine-tuned VLM for environmental hazard monitoring.
- *Advisor:* Prof. Lu Liang

Graduate Student Researcher, FIB-Lab, Dept. of Electronic Engineering

2024.09-2025.06

TSINGHUA UNIVERSITY

Beijing, China

- Led a project on LLM Spatial Intelligence, published on ACL Main 2025 (CCF-A).
- *Advisor:* Prof. Chen Gao, Prof. Jie Feng, Prof. Yong Li

Intern Planner & Architect

2024.10-2025.03

ARCHITECTURAL DESIGN & RESEARCH INSTITUTE OF TSINGHUA UNIVERSITY (THAD)

Beijing, China

- Zhejiang Wanli University Yuyao Campus Project (Constructed): Landscape planning and architectural design for two college buildings
- *Advisor:* Chief Architect Xiuquan Mo, Prof. Jinxi Chen

Intern Architect

2023.08-2024.06

TSINGHUA URBAN PLANNING & DESIGN INSTITUTE (THUPDI)

Beijing, China

- Beijing 1st Experimental Primary School Honglian Branch Renewal Project (Constructed)
- *Advisor:* Chief Architect Longlin Jin, Prof. Jinxi Chen

Intern Equity Researcher

2023.08-2024.06

KAIYUAN SECURITIES RESEARCH INSTITUTE

Shanghai, China

- Published an in-depth Report on the cosmetics industry and BeiTaiNi (SZ.300957)

TEACHING EXPERIENCE

UNIVERSITY OF CALIFORNIA, BERKELEY (First-Time GSI Prerequisites Fulfilled)

- [Fall 2025] ESPM 375 Teaching in Environmental Science, Policy, and Management
- [Fall 2025] Teaching Conference for First-Time GSIs
- [Fall 2025] GSI Professional Standards and Ethics Online Course

TSINGHUA UNIVERSITY

- [Fall 2023, Spring 2024, Fall 2024, Spring 2025] Undergraduate Thesis Studio (Teaching Assistant).
- [Fall 2023, Fall 2024] Urban Design Elements (Teaching Assistant).

TONGJI UNIVERSITY

- [Fall 2022] Building Mechanics and Structure (Tutor).

PROFESSIONAL SERVICES

PEER REVIEWER (13)

- GIScience & Remote Sensing(1); Building and Environment(4); Cities(2); Computational Urban Science(4); Architectural Engineering and Design Management(1); ICLR 2025

CONFERENCES & WORKSHOPS

- [2026] Annual Meeting of the Association of American Geographers (AAG), Session Organizer, "Exploring Spatial, Temporal, and Social Dimensions of Multiple Climatic Hazards", San Francisco, USA.
- [2024] Milan Summer Youth Festival, China-Italy Design and Innovation Hub, Tsinghua University, Student Volunteer Leader as "Interdisciplinary Design Innovation Scholars".

- [2023, 2024] The 3rd / 4th International Symposium of Urban Design Research and Education, Tsinghua University, Conference Assistant.

VOLUNTEER

- [2023- Now] Citipedia (#1 volunteer group promoting sustainable city & transportation in CN), Volunteer.
- [2021-2022] Student Union of College of Arch. & Urban Planning, Tongji University, President.
- [2021-2022] The “Dream Classroom” voluntary teaching + design + construction project, Tibet Lhatse Middle School, Xinjiang Huocheng Middle School, Tongji University, Leader.
- [2021] Archdaily (International Architecture Media), Volunteer Chinese Translator & Editor

PROFESSIONAL AFFILIATIONS

- [2022-2024] Architectural Society of China-Student Branch (ASC-SAC), Committee Member.
- [2026- Now] ACM Equity and Access in Algorithms, Mechanisms, and Optimization (ACM EAAMO), Urban Data Science & Equitable Cities Working Group Member.
- [2026- Now] ACM SIGSPATIAL Member
- [2022- Now] Architectural Society of China (ASC), Student Member.
- [2025- Now] American Association of Geographers (AAG), Student Member.
- [2025- Now] Association of Collegiate Schools of Planning (ACSP), Student Member.
- [2024- Now] International Association for China Planning (IACP), Student Member.
- [2024- Now] International Association of Chinese Professionals in GIS (CPGIS), Student Member.

CERTIFICATES & SKILLS

[2025] Certificate in Global Private Equity Research, Tsinghua University
[2023] Minor Program Certificate in Finance (27 credits), Fudan University
Language: English (Proficient); Chinese Mandarin (Native); Japanese, French (Basic)